2. Gradient Descent Method vs. Mini-Batch Gradient Descent Method vs. Stochastic Gradient Descent Method (SGD)

A. gradient descent method : compute the mean value of gradient descent for entire training examples, and then updating weights.
B. mini-batch gradient descent method : compute the mean value of gradient descent for a mini-batch ( a small set of entire training examples), and then updating weights.
C. stochastic gradient descent method : compute the value of gradient descent for a single training example, and updating weights.

A. gradient descent method : more accurate estimate of gradient descent , but more computation before updating weights ( less frequency in weight updated)
B. mini-batch gradient descent method : trade-off between accurate estimate of gradient descent and computation before updating weights ( frequency in weight updated)
C. stochastic gradient descent method : less accurate estimate of gradient descent, but less computation before updating weights ( more frequency in weight updated)

3. About Mini-Batch Gradient Descent Method in Matrix Multiplication:

each single pair training example is an array of inputs , and a mini-batch can be represented by a matrix because an array of an array of inputs becomes a matrix.
similarly, the weights for a single neuron is an array of weights , and all neurons in a layer can be represented by a matrix because an array of an array of weights becomes a matrix.

in the mini-batch gradient descent mechanism , multiplying the matrix of inputs (for a mini-batch ) and the matrix of weights ( for all activation functions of all neurons in a layer ) is then reduced to a single matrix multiplication.

it does lead to significant performance improvements on platforms that have highly efficient matrix multiplication implementations.